

Material Group

STEP 1 - Identify workpiece Material Group

DIN ISO 513	Material	Condition	Tensile Strength (Kpsi)	Hardness HB	Material Group # VDI 3323	Trade Names
P	Non-alloy steel, cast steel, free cutting steel	<0.25%C Annealed	61	125	1	1010, 1015, 1018, 1020, 1023, 1102, 1108, 1109, 1213, 1213, 1215
		>=0.25%C Annealed	94	190	2	1025, 1030, 1035, 1040, 1045, 1050, 1140, 1141, 1330
		<0.55%C Quenched and tempered	123	250	3	1025, 1030, 1035, 1040, 1045, 1050, 1140, 1141, 1330
		>=0.55%C Annealed	109	220	4	1055, 1060, 1070, 1080, 1151, W112
		Quenched and tempered	145	300	5	1055, 1060, 1151
	Low alloy steel and cast steel (Less than 5% of alloying elements)	Annealed	87	200	6	3135, 3435, 3440, 4130, 4140, 4150, 4320, 4340, 5015, 5060, 5120, 5132, 5140, 5160, 6150, 8620, 8640, 9254, 9255, 9262, L1, L2, L3, O1, O2, S1, P2, P3, P5, P20, W1, W2, W5, 51100, 52100,
		Quenched and tempered	135	275	7	
			145	300	8	
			174	350	9	
	High alloy steel, cast steel and tool steel	Annealed	99	200	10	A2, A3, A4, A6 A7, A10, D2, D3, D4, D5, D7, H11, H12, H13, H14, H17, H22, H23, S7, M1, M2, M3, M4, M6, M7, M10, M30, M42, T1, T2, T4, T5,
		Quenched and tempered	160	325	11	
M	Stainless steel and cast steel	Ferritic/martensitic	99	200	12	203, 303, 303Plus X, 303Pb, 409, 430, 430F, 440, 444
		Martensitic	119	240	13	410, 420, 431, 440A, 416
		Austenitic	87	180	14	304, 310, 316, 316L, 317, Nitronic 40, Carpenter 20, 13-8, 15-5, 17-4
K	Gray cast iron (GG)	Ferritic		180	15	ASTM A48 Classes 20, 25, SAEJ431c grades G1800, G2000, G2500
		Pearlitic		260	16	ASTM A48 Classes 30, 35, 40, 45, 50, SAEJ431c grades G3000, G3500
	Cast iron nodular (GGG)	Ferritic		160	17	60-40-18, 65-45-12
		Pearlitic		250	18	80-55-06, 100-70-03, 120-90-02
	Malleable cast iron	Ferritic		130	19	22010, 325110
		Pearlitic		230	20	40010, 50005, 70003, 90001
N	Aluminum - Wrought alloy	Not cureable		60	21	1000, 2011, 2014, 2124, 2024, 6061, 7075
		Cured		100	22	
	Aluminum - cast, alloyed	<=12% Si Not cureable		75	23	319, 383, 356, 413, 535, A280, A380, A413
		Cured		90	24	
		>12% Si High temp		130	25	
	Copper alloys	>1% Pb Free cutting		110	26	C23000, C36000, C83600, C93200, C93600
		Brass		90	27	C18200, C27200, C27700, C86500, Red Brass
		Electrolitic copper		100	28	B-148-52, C63000, C81500, C90700, C90800
	Non-metallic	Duroplastics, fiber plastics			29	
		Hard rubber			30	
S	High temp. alloys	Fe based Annealed		200	31	A286, Discaloy, Incoloy 800, Incoloy 801, Incoloy 802, N-155, W-545
				280	32	
		Ni or Co based Annealed		250	33	Astroloy, Hastelloy, C-272, IN-100, Inconel 625, Inconel 718, Inconel 750, Numonic, Rene, Udimet, Waspaoly, Haynes, Stellite, AR213, MP35N
				350	34	
				320	35	
	Titanium, Ti alloys		Rm 58		36	Grade 1, Grade 2, Grade 2H, Grade 3
		Alpha+beta alloys cured	Rm 152		37	Grade 5, Ti6AL4V, Ti6-4
H	Hardened steel	Hardened		55 HRC	38	Hardox 400, Hardox 500, W1, W210
		Hardened		60 HRC	39	HSS, 90 MnV8
	Chilled cast iron	Cast		400	40	Ni-Hard 1, Ni-Hard 2, Ni-Hard 4, A532, GX300
	Cast iron nodular	Hardened		55 HRC	41	